(a) Distances among stellar bodies do not change.
(b) Gravity of stellar bodies does not change.
(c) Light always travels in straight line.
(d) Speed of light is always same.

Ans. (d): Year is the distance a beam of light travels in a vacuum in one year. The reasons for measuring astronomical distances in light years are familiar units like kilometers and miles are absurdly small for describing the vastness of the cosmos and the speed of light is both finite and constant with a value of approximately 2.998 x 108 m/s.
Hence option (d) is the correct answer.

ENVIRONMENT AND ECOLOGY

70. Why is there a concern about copper smelting plants:
1. They may release lethal quantities of carbon monoxide into environment.
2. The copper slag can cause the leaching of some heavy metals into environment.
3. They may release sulphur dioxide as a pollutant.

Select the correct answer using the code given below.
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (b): Every smelting plant has its own oxygen production unit, to feed in the smelting process as well as reduce the exhaust of gases. So, carbon monoxide although produced in the smelting process would get converted to carbon dioxide if the plant is functioning properly and if there are any issues with the oxygen-producing unit, the smelting process will slow down. So the release of a lethal amount of carbon mono oxide. So, statement 1 is correct.
The other byproduct of smelting is slag, the waste matter separated from metals. This slag may leach heavy metals (arsenic, cadmium, lead or mercury depending on the composition of the ore) into ground water reservoirs. Water which has a high heavy metal content is very toxic to humans. Slag may also increase the concentration of other, less harmful salts in water resources, which may change the taste of water. So, statement 2 is correct.
Most copper ores are sulphur-based and smelting releases sulphur dioxide, an air pollutant known to have many harmful effects. So, statement 3 is correct.
Therefore, the correct answer is (b).

71. With reference to furnace oil, consider the following statements :
1. It is a product of oil refineries.
2. Some industries use it to generate power.
3. Its use causes sulphur emission into environment.

Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d): Statement 1 is correct: Furnace oil or fuel oil is a dark viscous residual fuel obtained by blending mainly heavier components from crude distillation unit, short residue and clarified oil from fluidized catalytic cracker unit.
Statement 2 is correct: It is used primarily for steam boilers in power plants, aboard ships, and in industrial plants.
Statement 3 is correct: Sulphur contained in fuel causes emissions of sulphur dioxide (SO_2) and also contributes to the formation of secondary particulate matter (PM).

72. In the nature, which of the following is/are most likely to be found surviving on a surface without soil?
1. Fern 2. Lichen
3. Moss 4. Mushroom

Select the correct answer using the code given below
(a) 1 and 4 only (b) 2 only
(c) 2 and 3 (d) 1, 3 and 4

Ans. (c): Ferns require indirect sunlight, moist soil, and a humid atmosphere. Lichens occur from sea level to high alpine elevations, in many environmental conditions, and can grow on almost on any surface. Mosses are non-flowering plants which produce spores and have stems and leaves, but don't have true roots. The individual plants are usually composed of simple leaves that are generally only one cell thick, attached to a stem that may be branched or un-branched and has only a limited role in conducting water and nutrients. A mushroom or toadstool is the fleshy, spore-bearing fruiting body of a fungus, typically produced above ground, on soil, or on its food source.

73. Which one of the following is used in preparing a natural mosquito repellent?
(a) Congress grass (b) Elephant grass
(c) Lemongrass (d) Nut grass

Ans. (c): Lemongrass used in preparing a natural mosquito repellent found in candles, sprays and lotions.
Hence, option (c) is the correct answer.

74. Consider the following kinds of organisms :
1. Copepods 2. Cyanobacteria
3. Diatoms 4. Foraminifera

Which of the above are primary producers in the food chains of oceans?
(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 1 and 4

Ans. (b) : Cyanobacteria and Diatoms are the primary producers in the food chains of oceans and copepods and foraminifera are not the primary producers in the food chains of oceans.
Hence, option (b) is the correct answer.

75. Consider the following animals :
1. Hedgehog
2. Marmot
3. Pangolin

To reduce the chance of being captured by predators, which of the above organisms rolls up/roll up and protects/protect its/their vulnerable parts?
(a) 1 and 2 (b) 2 only
(c) 3 only (d) 1 and 3

Ans. (d): Hedgehogs have spikes on their back, which help to deter any threat when they roll up & protect their vulnerable parts.
Marmots build elaborate burrows for dinning and protection from predators. They have no rolling up mechanism like other 2.
Pangolins have hard scales on their back which help to protect its vulnerable parts when it rolls up.
Therefore, the correct answer is (d).

76. With reference to the 'New York Declaration on Forests', which of the following statements are correct?
1. It was first endorsed at the United Nations Climate Summit in 2014.
2. It endorses a global timeline to end the loss of forests.
3. It is a legally binding international declaration.
4. It is endorsed by governments, big companies and indigenous communities.
5. India was one of the signatories at its inception.

Select the correct answer using the code given below.
(a) 1, 2 and 4 (b) 1, 3 and 5
(c) 3 and 4 (d) 2 and 5

Ans. (a): As per the given statements, statements 1, 2 and 4 are correct in reference to the 'New York Declaration on Forests'. The New York Declaration on forests (NYDF) is a political declaration calling for global action to protect and restore forests. It offers a common, multi-stakeholder framework for forest action, consolidating various initiatives and objectives that drive forest protection, restoration and sustainable use.
Hence option (a) is the correct answer.

77. Magnetite particles, suspected to cause neurodegenerative problems, are generated as environmental pollutants from which of the following?
1. Brakes of motor vehicles
2. Engines of motor vehicles
3. Microwave stoves within homes
4. Power plants
5. Telephone lines

Select the correct answer using the code given below.
(a) 1, 2, 3 and 5 only (b) 1, 2 and 4 only
(c) 3, 4 and 5 only (d) 1, 2, 3, 4 and 5

Ans. (d): Magnetite can have potentially large impacts on the brain due to its unique combination of redox activity, surface charge and strongly magnetic behavior. Airborne magnetite pollution particles < ~200 nm in size can access the brain directly via the olfactory and/or trigeminal nerves, bypassing the blood -brain barrier. These tiny particles, each less than a thousandth the width of a human hair, are generated by a wide variety of processes, both natural and fabricated, including forest fires and wood -burning stoves, road traffic pollution, and many high -temperature industrial processes.
In addition to occupational settings (including, for example, exposure to printer toner powders), higher concentrations of magnetite pollution nano particles may arise in the indoor environment from open fires or poorly-sealed stoves used for cooking and/or heating, and in the outdoor environment from vehicle (tailpipe and brake wear) and/or industrial PM sources.
So option (d) is the correct answer.

78. Which one of the following is a filter feeder?
(a) Catfish (b) Octopus
(c) Oyster (d) Pelican

Ans. (c): Filter feeder is an animal that obtains its good by filtering organic matter or minute organisms from a current of water that passes through some part of its system, Oysters are natural filter feeders.
Hence, option (c) is the correct answer.

79. In case of which one of the following biogeochemical cycles, the weathering of rocks is the main source of release of nutrient to enter the cycle?
(a) Carbon cycle (b) Nitrogen cycle
(c) Phosphorus cycle (d) Sulphur cycle

Ans. (c): Phosphorus is tied up in rock and sedimentary deposits from which it is released by weathering, leaching and mining.
Hence, option (c) is the correct answer.

80. Which of the following are detritivores?
1. Earthworms 2. Jellyfish
3. Millipedes 4. Seahorses
5. Woodlice

Select the correct answer using the code given below.
(a) 1, 2 and 4 only (b) 2, 3, 4 and 5 only
(c) 1, 3 and 5 only (d) 1, 2, 3, 4 and 5

Ans. (c): An animal that feeds on detritus. Examples of detritivores are earthworms, blowflies, maggots, millipedes, dung beetles, fiddler crabs, sea cucumbers and woodlice.
Detritivores play an important role in the breakdown of organic matter from decomposing animals and plants
A seahorse gains energy by eating other live organisms, meaning it is a carnivore. A seahorse is not a decomposer because decomposers gain energy from breaking down dead organisms.
Jellyfish are carnivores and excellent predators. They sting with tentacles to subdue small aquatic fish and eat the eggs and invertebrates that stick to their tentacles
Therefore, the correct answer is (c)..